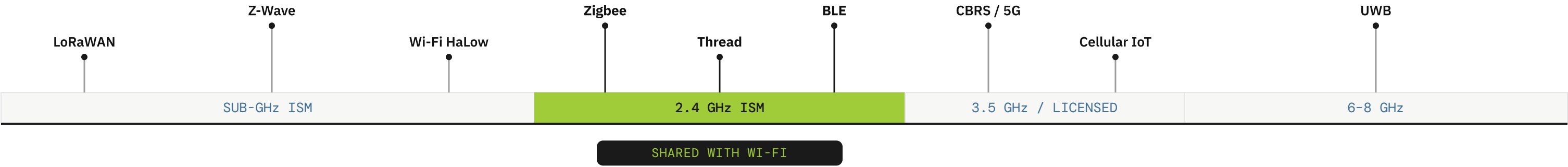


The radio landscape a WLAN pro lives in

Five systems share your 2.4 GHz air; the rest live in sub-GHz or licensed spectrum and never touch it. Where each radio sits, and whether it contends for the airtime you plan.



SYSTEM	BAND(S)	RANGE	DATA RATE	POWER	TOPOLOGY	TYPICAL USE	SHARES 2.4 GHz?
LoRaWAN LoRa Alliance	Sub-GHz (US 902–928 / EU 863–870 MHz)	2–15 km	0.3–50 kbps	Very low (5–10 yr)	Star-of-stars	Metering, ag, city sensors	✔ No (sub-GHz)
Zigbee CSA	2.4 GHz (868/915 MHz opt.)	10–100 m / hop	250 kbps	Very low (yrs)	Self-healing mesh	Lighting, building automation	⚠ Yes
Thread Thread Group / CSA	2.4 GHz 802.15.4	10–100 m / hop	250 kbps	Very low (yrs)	Self-healing mesh (IPv6)	Matter device transport	⚠ Yes
BLE Bluetooth SIG	2.4 GHz (40 ch, AFH)	10 m–~1 km (Coded)	125 kbps–2 Mbps	Very low (mo–yrs)	P2P / star / beacon / mesh	Wearables, beacons, RTLS	⚠ Yes
Z-Wave Z-Wave Alliance	Sub-GHz (US 908.42 / EU 868.42 MHz)	30 m–~400 m (LR)	9.6–100 kbps	Low (yrs)	Mesh / LR star	Home automation	✔ No (sub-GHz)
UWB IEEE 802.15.4z	3.1–10.6 GHz	~50–200 m	Ranging, not data	Moderate (bursts)	P2P TWR / TDoA anchors	cm-precise location, keyless	✔ No (6–8 GHz)
Cellular IoT 3GPP (NB-IoT / LTE-M)	Licensed carrier, sub-6 GHz	km (deep-coverage)	20 kbps–1 Mbps	Very low (~10 yr)	Star (to base station)	Wide-area sensors, moving assets	✔ No (licensed)
Wi-Fi HaLow IEEE 802.11ah	Sub-GHz (US 902–928 / EU 863–868 MHz)	~1 km	150 kbps–78 Mbps	Low (TWT)	Star (~8191 sta)	Long-range IoT, cameras	✔ No (sub-GHz)
CBRS / private 5G FCC (n48 / B48)	3.55–3.70 GHz	Campus	Mbps–Gbps	Wired / powered	Star (SAS-coordinated)	Enterprise private cellular	✔ No (3.5 GHz)

Matter is not a radio - it is an application layer that rides Thread and Wi-Fi, commissioned over BLE. And **CBRS does not interfere with your Wi-Fi**; on 3.5 GHz it competes for the job, not the airtime. Range, rate, and battery are real-world envelopes, not hard specs.